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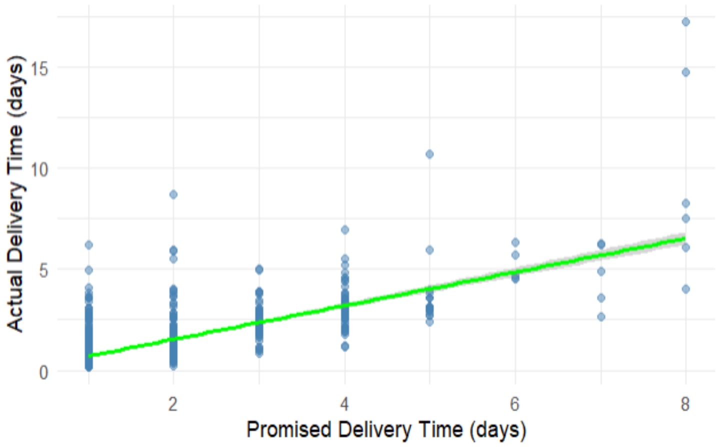
Predicting E-Commerce Delivery Time Using Linear Regression

Overview

A random sample of 1,000 transactions from an e-commerce platform's March 2018 data was analysed using the seeding technique. The dataset includes four variables such as *promised_delivery_time*, *actual_delivery_time, order_type*, and *same_region*. A multiple linear regression model was developed to predict *actual_delivery_time* using *promise* and *same_region* as predictor variables. The former reflects service target, while the latter captures delivery efficiency, as the same location deliveries are typically faster.

Calculated Pearson correlation coefficient of 0.72 between *promise* and a*ctual_delivery_time* indicates a strong positive linear relationship, showing that longer promise time leads to longer delivery times. This relationship is visually shown in *Figure 1.1*.

Fig 1.1: *Relationship between promised, actual delivery time*



Model Development and Evaluation

The following linear regression model was built to predict actual delivery time based on promised delivery time and whether the delivery occurred within the same region.

Table 1: *Linear regression results for predicting actual delivery time*

Variable	Estimate	Std. Error	t value	Pr(>|t|)	Significance
Intercept	-0.46243	0.15423	-2.998	0.00278	**
Promise	0.85531	0.02737	31.248	<2e-16	***
same_region	0.31589	0.13284	2.378	0.0176	*
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					
Residual standard error: 0.9255 on 997 degrees of freedom					
Multiple R-squared: 0.5224, Adjusted R-squared: 0.5214					
F-statistic: 545.2 on 2 and 997 DF, **p-value: < 2.2e-16**					

As shown in *Table 1*, the selected variables are statistically significant, and the overall model also highlights a strong statistical significance. This model achieves an **R^2 of 52.2%,** indicating that the two variables explain just over half of the variance in actual delivery time.

Accordingly, a multiple linear regression model was built with the following equation:

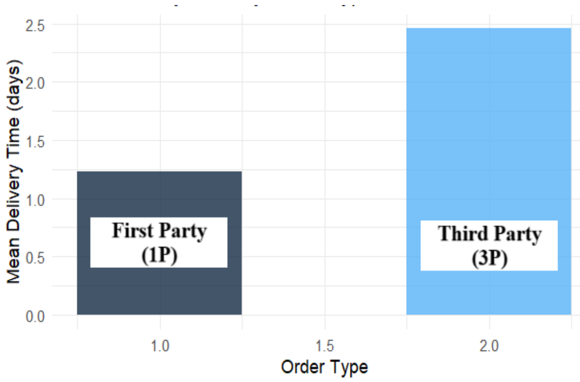
$\hat{y} = \beta_0 \text{ (Intercept)} + \beta_1 \text{ (coefficient 1)} + \beta_2 \text{ (coefficient 2)}$

actual_delivery_time = -0.462 + 0.855 × promise + 0.316 × same_region

For example, a 5-day promised delivery for a same-region order predicts ~4.13 days actual delivery, while a 2-day cross-region order predicts ~1.24 days.

Additional Insight on Order_type

Fig 1.2: *Mean delivery time by order type*



Although not included in the regression model due to low statistical significance, *order_type* provided meaningful descriptive insights. On average, first-party (1P) orders are delivered in approximately one day, whereas third-party (3P) orders take around three days, highlighting notable operational differences as illustrated in *Figure 1.2*.

Business Recommendation

This model offers a data-driven approach to predict delivery time, highlighting the importance of promised time and regional closeness in logistics planning. Secondly, longer promised times often lead to slower handling, which can be improved by treating all orders equally urgent. The delivery gap between first-party and third-party orders also suggests a need to optimise third-party operations. These insights can improve delivery forecasting, enhance customer communication, and strengthen overall service performance.